40 A neutron is composed of one up (u) quark and two down (d) quarks. When a neutron decays to a proton, a beta-particle is emitted.

What is the change in the quark structure of the neutron due to the emission of the beta-particle?

(The symbol for a neutrino is ν_e and for an antineutrino is $\overline{\nu_e}$.)

A $d \rightarrow u + \beta^- + \nu_e$

B $d \rightarrow u + \beta^- + \overline{\nu_e}$

C $u \rightarrow d + \beta^+ + \nu_e$

D $u \rightarrow d + \beta^+ + \overline{\nu_e}$